GONetView

A separate web application for drawing Gene Ontology term networks, and what does and does not pass between it and this tool.

AT A GLANCE

Where	External Tools → GONetView. In the DataViewer it is the only page of that section.
What it is	A web application in its own right, maintained separately from this tool and published with a DOI and an open source repository. It is not part of the analysis, and nothing passes between the two on its own.
Sends	Nothing. This page holds a description, a picture and three links; it reads no dataset and transmits no result.
Needs	A gene list, which you take there yourself.
Returns	Nothing to this session. What you make there you save there.
New	Not covered by any previous manual.

1 What the page is

It is a card, not a tool. The page describes GONetView — an application for visualising Gene Ontology term networks and for seeing how genes connect to biological processes, molecular functions and cellular components — lists what it is good for, shows a picture of it, and offers three links: **Open GONetView**, **DOI** and **GitHub**. The picture is a link to the same application. Under them is the citation to use if a figure made there ends up in a paper, with the software's version and its DOI.

Each link opens in a new browser tab and tells the destination nothing about where it was opened from. Nothing else happens: no request carries your data, because the page has none — it does not read the uploaded datasets, the annotation or any result.

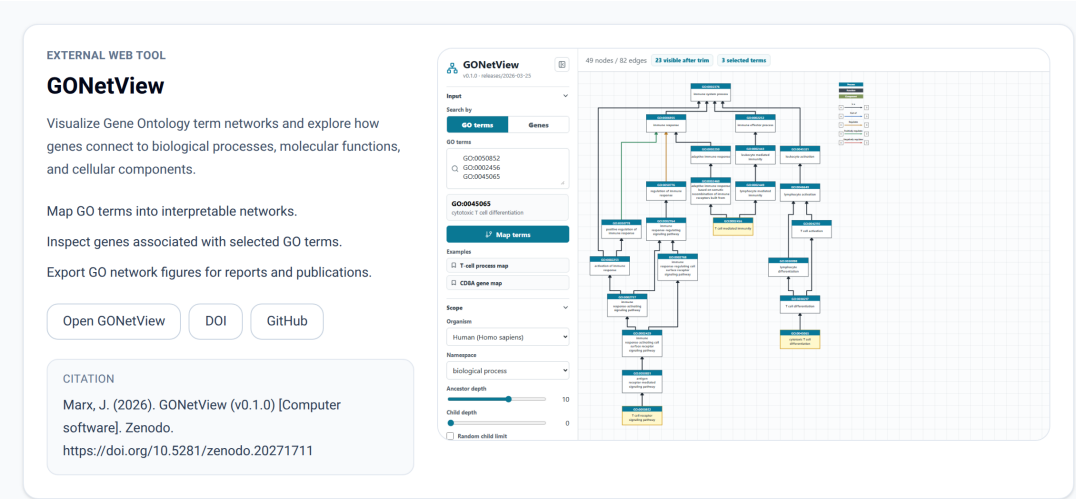


Figure 1.1. The GONetView page: what the tool does, the three links out to it, the citation to use, and a picture of the application itself — which is also a link. Nothing on this page sends anything anywhere; opening a link is what starts a session with the other application.

> **NOTE**
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> The line between the two applications runs through your browser, not through the tool. Opening the link starts a session with somebody else's application: what you paste into it is sent to whoever hosts it, under their terms, and this tool is neither party to that nor aware of it.

2 Taking a list there

Nothing is handed over automatically, so the list has to be carried.

1. Produce the genes in the tool. The usual source is Statistics → ORA, which downloads its up and down gene lists as plain text and its full enrichment table — terms, sizes, overlaps, p-values and the intersecting genes themselves — as CSV.
2. Open GONetView from this page. The tool goes on running behind it; nothing in the session changes, then or later.
3. Paste or upload the list there, and work in that application.
4. Save what you want to keep from it. A figure made there is a file on your disk; it does not come back into this analysis, and no page of this tool will show it.

> **EASY TO GET WRONG**
>
> The enrichment and the network answer different questions. ORA tests whether a term is over-represented in one list against one background, with a one-sided hypergeometric test and a Benjamini–Hochberg correction, and its p-values depend on which background was chosen. A network drawing of terms and their genes does no test at all: an impressive-looking network of terms can be drawn from a gene list that is not enriched for any of them. Take the statistics from the ORA table and the picture from the network, never the other way round.

3 If it cannot be reached

The page itself is part of the tool and always opens, with its text, links and citation intact. Two things depend on the outside:

- The picture is fetched from the repository that hosts GONetView. Without a network, or behind a firewall that blocks that host, the frame stays empty. Nothing else on the page depends on it.
- The links are handled by the browser. If the application cannot be reached, the browser reports that, in its own words, in the new tab. The tool learns nothing about it and reports nothing.

No analysis, no plot and no download in this tool depends on GONetView being reachable. An instance installed on a machine with no internet access works exactly as it does otherwise, minus the picture and the links.

GONetView takes Gene Ontology terms, GO identifiers or a gene list.

Carrying the list across by hand is the intended route: GONetView is a separate tool, and this page is a pointer to it rather than an integration.